Assignment - 02

Q.1. What are the different protocols you observe at the following layers of the protocol stack?

A.) Application layer (B.) Transport layer (C.) Network layer

A.) Application layer:
- MDNS
- NBNS
- DHCP
- DNS
- HTTP

(B.) Transport layer:
- UDP
- TCP

(C.) Network layer:
- ARP

Q.2. What is the total amount of data being received for the following two cases?

A.) When you access http://neverssl.com/

B.) When you access http://www.cornell.edu *any

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ip.addr==10.85.18.54

No.	Time	Source	Destination	Protocol	Length	Info
5	0.110182367	10.85.18.54	255.255.255.255	UDP	870	60661 → 29810 Len=826

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ip.addr==10.85.23.221

No.	Time	Source	Destination	Protocol	Length	Info
8	0.211023302	10.85.23.221	255.255.255.255	NBNS	94	Name query NB *<00><00><00><00><00><00><00><00><00><00><00><00><00><00><00>
35	0.819896494	10.85.23.221	224.0.0.251	MDNS	84	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
36	0.819897269	10.85.23.221	224.0.0.251	MDNS	167	Standard query 0x0000 PTR _ftp._tcp.local, "QM" question PTR _smb._tcp.local, "QM" question PTR _nfs._tcp.local, "QM" questio…
431	12.395436067	10.85.23.221	255.255.255.255	NBNS	94	Name query NB *<00><00><00><00><00><00><00><00><00><00><00><00><00><00><00>
551	15.772669002	10.85.23.221	224.0.0.251	MDNS	167	Standard query 0x0000 PTR _ftp._tcp.local, "QM" question PTR _smb._tcp.local, "QM" question PTR _nfs._tcp.local, "QM" questio…
552	15.772669523	10.85.23.221	224.0.0.251	MDNS	84	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
1043	31.031731555	10.85.23.221	255.255.255.255	NBNS	94	Name query NB *<00><00><00><00><00><00><00><00><00><00><00><00><00><00><00>
1047	31.242557825	10.85.23.221	224.0.0.251	MDNS	167	Standard query 0x0000 PTR _ftp._tcp.local, "QM" question PTR _smb._tcp.local, "QM" question PTR _nfs._tcp.local, "QM" questio…

B.) when you access http://www.cornell.edu

Q.3. Answer the following when you access the site
http://neversal.com/

A.) How many HTTP GET request do you observe? List down the GET requests.

→ 1 HTTP GET request observe

B.) what version of HTTP the server is running?

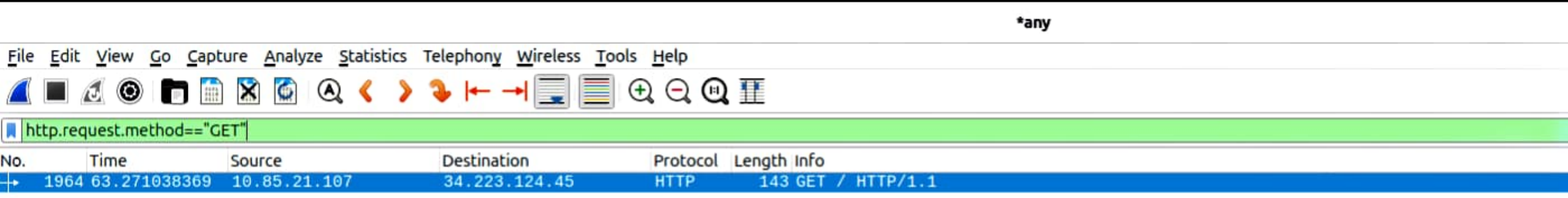
→ Time = 63.2710 38369
Source = 10.85.21.107
Destination = 34.223.124.45
Protocol = HTTP
length/Info = 143 GET /HTTP/1.1

c.) For each of the HTTP GET request, find out:
i) The total number of TCP segment being received
→ 3 TCP segment

ii) The total amount of data being received in the corresponding HTTP response message
→ Tap segment length = 0

D.) The identify TCP three ways handshake used to establish the connection with the http://neversal.com. Specify SYN, SYN-ACK, ACK packets and write the sequence number of each of the three accesses.

→ SYN sequence NO = 0
SYN ACK sequence NO = 0
ACK sequence Number = 1


*any
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http.request.method=="GET"
No. Time Source Destination Protocol Length Info
1964 63.271038369 10.85.21.107 34.223.124.45 HTTP 143 GET / HTTP/1.1

Apply a display filter … <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1107	29.089349700	fe80::e45f:1bff:fe4…	ff02::fb	MDNS	506	Standard query response 0x0000 PTR, cache flush Android_0DX7M5NB.local A, cache flush 10.85.19.118 PTR, cache flush And…
1108	29.089360082	8a:26:d0:f4:f4:72	Broadcast	ARP	60	Gratuitous ARP for 10.85.22.163 (Reply)
1109	29.191440103	10.85.17.10	255.255.255.255	UDP	868	44246 → 29810 Len=826
1110	29.191448414	10.85.22.43	255.255.255.255	UDP	867	47698 → 29810 Len=825
1111	29.191467252	b0:19:21:74:04:80	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.17.10
1112	29.191467837	b0:19:21:74:0d:0a	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.22.43
1113	29.290422661	ASIXElectron_35:2d:…	Broadcast	ARP	60	Who has 10.85.23.1? Tell 10.85.16.1
1114	29.290423963	b0:19:21:74:0d:48	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.22.105
1115	29.290424107	b0:19:21:d4:8e:fa	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.16.145
1116	29.292714229	34.223.124.45	10.85.17.18	TCP	66	80 → 34366 [ACK] Seq=1 Ack=77 Win=29312 Len=0 TSval=3422388319 TSecr=2905658291
1117	29.292715891	34.223.124.45	10.85.17.18	HTTP	4327	HTTP/1.1 200 OK (text/html)
1118	29.292843925	10.85.17.18	34.223.124.45	TCP	66	34366 → 80 [ACK] Seq=77 Ack=4262 Win=60416 Len=0 TSval=2905658597 TSecr=3422388320
1119	29.294132074	10.85.17.18	34.223.124.45	TCP	66	34366 → 80 [FIN, ACK] Seq=77 Ack=4262 Win=60416 Len=0 TSval=2905658598 TSecr=3422388320
1120	29.399267293	10.85.17.205	255.255.255.255	UDP	868	57137 → 29810 Len=826
1121	29.399265793	10.85.18.54	255.255.255.255	UDP	871	33504 → 29810 Len=829
1122	29.399308352	b0:19:21:74:06:6e	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.18.54
1123	29.399308748	b0:19:21:d4:90:f8	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.17.205
1124	29.399308803	ASIXElectron_35:2d:…	Broadcast	ARP	60	Who has 10.85.18.83? Tell 10.85.16.1
1125	29.399308858	ASIXElectron_35:2d:…	Broadcast	ARP	60	Who has 10.85.20.231? Tell 10.85.16.1
1126	29.496981642	10.85.22.234	224.0.0.251	MDNS	82	Standard query 0x0001 PTR _googlecast._tcp.local, "QU" question
1127	29.496982349	fe80::2c6d:75ff:fe1…	ff02::fb	MDNS	102	Standard query 0x0001 PTR _googlecast._tcp.local, "QU" question
1128	29.496982474	10.85.22.234	224.0.0.251	MDNS	136	Standard query 0x0001 PTR _googlecast._tcp.local, "QU" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B27C5ED._sub._go…

```
› Frame 1117: 4327 bytes on wire (34616 bits), 4327 bytes captured (34616 bits) on interface wlp0s20f3, i…
› Ethernet II, Src: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0), Dst: 44:38:e8:60:4d:a2 (44:38:e8:60:4d:a2)
› Internet Protocol Version 4, Src: 34.223.124.45, Dst: 10.85.17.18
⌄ Transmission Control Protocol, Src Port: 80, Dst Port: 34366, Seq: 1, Ack: 77, Len: 4261
    Source Port: 80
    Destination Port: 34366
    [Stream index: 0]
  › [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 4261]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 2373345713
    [Next Sequence Number: 4262    (relative sequence number)]
    Acknowledgment Number: 77    (relative ack number)
    Acknowledgment number (raw): 2743240070
    1000 .... = Header Length: 32 bytes (8)
  › Flags: 0x018 (PSH, ACK)
    Window: 229
    [Calculated window size: 29312]
    [Window size scaling factor: 128]
    Checksum: 0xcb3e [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  › Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  › [Timestamps]
  › [SEQ/ACK analysis]
    TCP payload (4261 bytes)
› Hypertext Transfer Protocol
› Line-based text data: text/html (131 lines)
```



```
0040  d3 b3 48 54 54 50 2f 31  2e 31 20 32 30 30 20 4f   ··HTTP/1 .1 200 O
0050  4b 0d 0a 44 61 74 65 3a  20 54 68 75 2c 20 30 39   K··Date:  Thu, 09
0060  20 4a 75 6c 20 32 30 32  36 20 30 37 3a 33 32 3a    Jul 202 6 07:32:
0070  33 34 20 47 4d 54 0d 0a  53 65 72 76 65 72 3a 20   34 GMT·· Server: 
0080  41 70 61 63 68 65 2f 32  2e 34 2e 36 36 20 28 29   Apache/2 .4.66 ()
0090  0d 0a 55 70 67 72 61 64  65 3a 20 68 32 2c 68 32   ··Upgrad e: h2,h2
00a0  63 0d 0a 43 6f 6e 6e 65  63 74 69 6f 6e 3a 20 55   c··Conne ction: U
00b0  70 67 72 61 64 65 0d 0a  4c 61 73 74 2d 4d 6f 64   pgrade·· Last-Mod
00c0  69 66 69 65 64 3a 20 57  65 64 2c 20 32 39 20 4a   ified: W ed, 29 J
00d0  75 6e 20 32 30 32 32 20  30 30 3a 32 33 3a 33 33   un 2022  00:23:33
00e0  20 47 4d 54 0d 0a 45 54  61 67 3a 20 22 66 37 39    GMT··ET ag: "f79
00f0  2d 35 65 32 38 62 32 39  64 33 38 65 39 33 22 0d   -5e28b29 d38e93"·
0100  0a 41 63 63 65 70 74 2d  52 61 6e 67 65 73 3a 20   ·Accept- Ranges: 
0110  62 79 74 65 73 0d 0a 43  6f 6e 74 65 6e 74 2d 4c   bytes··C ontent-L
0120  65 6e 67 74 68 3a 20 33  39 36 31 0d 0a 56 61 72   ength: 3 961··Var
0130  79 3a 20 41 63 63 65 70  74 2d 45 6e 63 6f 64 69   y: Accep t-Encodi
0140  6e 67 0d 0a 43 6f 6e 74  65 6e 74 2d 54 79 70 65   ng··Cont ent-Type
0150  3a 20 74 65 78 74 2f 68  74 6d 6c 3b 20 63 68 61   : text/h tml; cha
0160  72 73 65 74 3d 55 54 46  2d 38 0d 0a 0d 0a 3c 68   rset=UTF -8····<h
0170  74 6d 6c 3e 0a 09 3c 68  65 61 64 3e 0a 09 09 3c   tml>··<h ead>···<
0180  74 69 74 6c 65 3e 4e 65  76 65 72 53 53 4c 20 2d   title>Ne verSSL -
0190  20 43 6f 6e 6e 65 63 74  69 6e 67 20 2e 2e 2e 20    Connect ing ... 
01a0  3c 2f 74 69 74 6c 65 3e  0a 09 09 3c 73 74 79 6c   </title> ···<styl
01b0  65 3e 0a 09 09 62 6f 64  79 20 7b 0a 09 09 09 66   e>···bod y {····f
01c0  6f 6e 74 2d 66 61 6d 69  6c 79 3a 20 4d 6f 6e 74   ont-fami ly: Mont
01d0  73 65 72 72 61 74 2c 20  68 65 6c 76 65 74 69 63   serrat,  helvetic
01e0  61 2c 20 61 72 69 61 6c  2c 20 73 61 6e 73 2d 73   a, arial , sans-s
01f0  65 72 69 66 3b 0a 09 09  09 66 6f 6e 74 2d 73 69   erif;··· ·font-si
0200  7a 65 3a 20 31 36 78 3b  0a 09 09 09 63 6f 6c 6f   ze: 16x; ····colo
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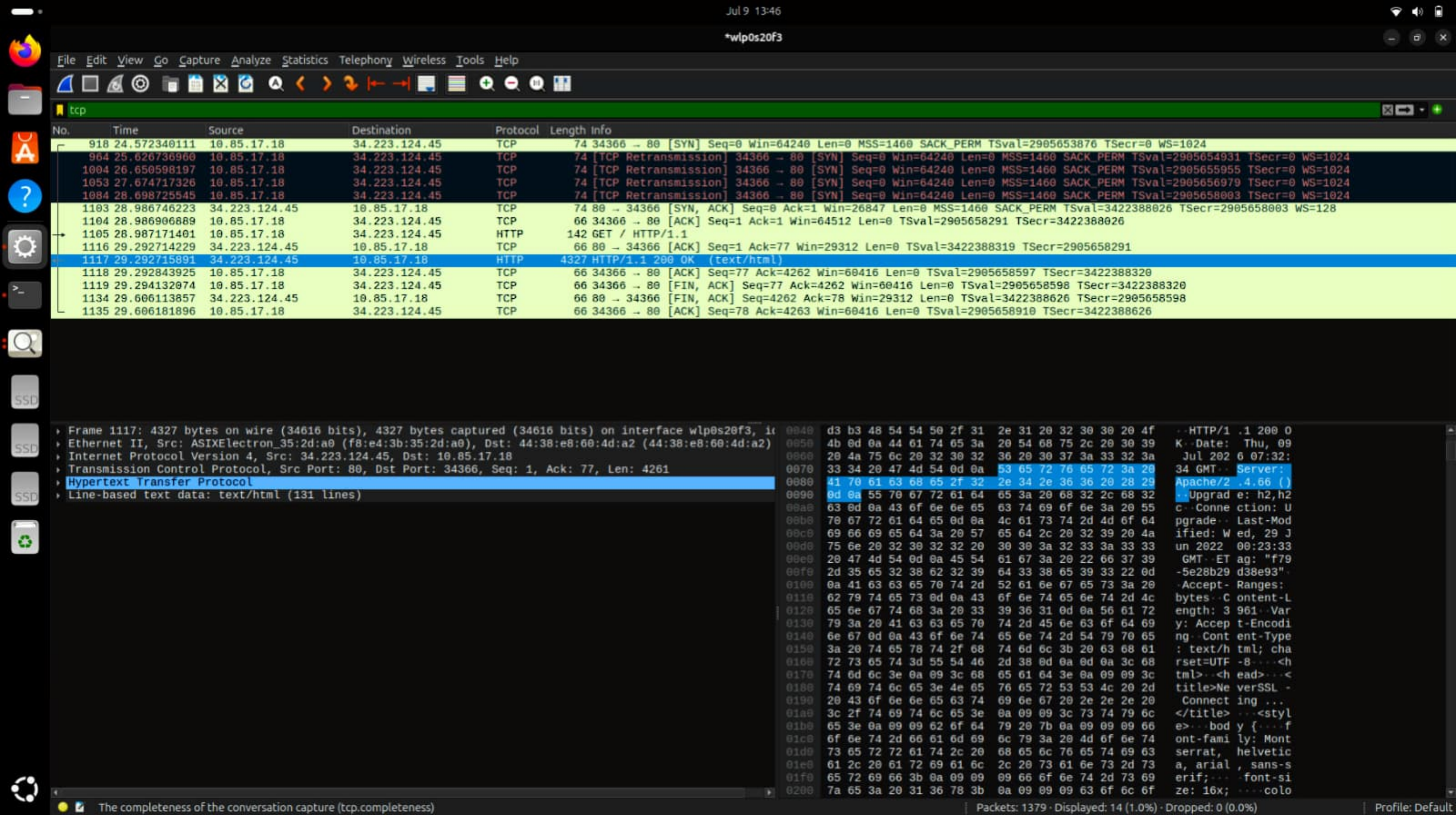
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*wlp0s20f3
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
918 24.572340111 10.85.17.18 34.223.124.45 TCP 74 34366 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2905653876 TSecr=0 WS=1024
964 25.626736960 10.85.17.18 34.223.124.45 TCP 74 [TCP Retransmission] 34366 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2905654931 TSecr=0 WS=1024
1004 26.650598197 10.85.17.18 34.223.124.45 TCP 74 [TCP Retransmission] 34366 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2905655955 TSecr=0 WS=1024
1053 27.674717326 10.85.17.18 34.223.124.45 TCP 74 [TCP Retransmission] 34366 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2905656979 TSecr=0 WS=1024
1084 28.698725545 10.85.17.18 34.223.124.45 TCP 74 [TCP Retransmission] 34366 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2905658003 TSecr=0 WS=1024
1103 28.986746223 34.223.124.45 10.85.17.18 TCP 74 80 → 34366 [SYN, ACK] Seq=0 Ack=1 Win=26847 Len=0 MSS=1460 SACK_PERM TSval=3422388026 TSecr=2905658003 WS=128
1104 28.986906889 10.85.17.18 34.223.124.45 TCP 66 34366 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2905658291 TSecr=3422388026
1105 28.987171401 10.85.17.18 34.223.124.45 HTTP 142 GET / HTTP/1.1
1116 29.292714229 34.223.124.45 10.85.17.18 TCP 66 80 → 34366 [ACK] Seq=1 Ack=77 Win=29312 Len=0 TSval=3422388319 TSecr=2905658291
1117 29.292715891 34.223.124.45 10.85.17.18 HTTP 4327 HTTP/1.1 200 OK (text/html)
1118 29.292843925 10.85.17.18 34.223.124.45 TCP 66 34366 → 80 [ACK] Seq=77 Ack=4262 Win=60416 Len=0 TSval=2905658597 TSecr=3422388320
1119 29.294132074 10.85.17.18 34.223.124.45 TCP 66 34366 → 80 [FIN, ACK] Seq=77 Ack=4262 Win=60416 Len=0 TSval=2905658598 TSecr=3422388320
1134 29.606113857 34.223.124.45 10.85.17.18 TCP 66 80 → 34366 [FIN, ACK] Seq=4262 Ack=78 Win=29312 Len=0 TSval=3422388626 TSecr=2905658598
1135 29.606181896 10.85.17.18 34.223.124.45 TCP 66 34366 → 80 [ACK] Seq=78 Ack=4263 Win=60416 Len=0 TSval=2905658910 TSecr=3422388626
Frame 1117: 4327 bytes on wire (34616 bits), 4327 bytes captured (34616 bits) on interface wlp0s20f3, id
Ethernet II, Src: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0), Dst: 44:38:e8:60:4d:a2 (44:38:e8:60:4d:a2)
Internet Protocol Version 4, Src: 34.223.124.45, Dst: 10.85.17.18
Transmission Control Protocol, Src Port: 80, Dst Port: 34366, Seq: 1, Ack: 77, Len: 4261
Hypertext Transfer Protocol
Line-based text data: text/html (131 lines)
The completeness of the conversation capture (tcp.completeness)
Packets: 1379 · Displayed: 14 (1.0%) · Dropped: 0 (0.0%)
Profile: Default